

GROUP NO 9.

APPLET CODE FOR RACING CAR GAME.

APPLET CODE

```
import java.applet.Applet;
import java.awt.*;
import java.awt.event.*;

public class CarRacingGame extends Applet implements KeyListener, Runnable {
    private final int WINDOW_WIDTH = 800;
    private final int WINDOW_HEIGHT = 600;
    private final int CAR_WIDTH = 50;
    private final int CAR_HEIGHT = 100;
    private final int TRACK_WIDTH = 600;
    private final int TRACK_HEIGHT = 400;
    private final int FINISH_LINE = TRACK_WIDTH - 100;

    private int redCarX, redCarY, blueCarX, blueCarY;
    private boolean isRedCarMovingLeft, isRedCarMovingRight, isBlueCarMovingLeft,
isBlueCarMovingRight;
    private boolean isGameRunning;
    private boolean isGameFinished;
    private String winnerMessage;

    public void init() {
        setSize(WINDOW_WIDTH, WINDOW_HEIGHT);
        addKeyListener(this);
        redCarX = 100;
        redCarY = WINDOW_HEIGHT - CAR_HEIGHT - 50;
        blueCarX = WINDOW_WIDTH - CAR_WIDTH - 100;
        blueCarY = WINDOW_HEIGHT - CAR_HEIGHT - 50;
        isGameRunning = true;
        isGameFinished = false;
        new Thread(this).start();
    }

    public void run() {
        while (isGameRunning) {
            updateCarPositions();
            checkFinishLine();
            repaint();
            try {
                Thread.sleep(20);
            } catch (InterruptedException e) {
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        e.printStackTrace();
    }
}

private void updateCarPositions() {
    if (isRedCarMovingLeft && redCarX > 0) {
        redCarX -= 5;
    } else if (isRedCarMovingRight && redCarX < TRACK_WIDTH - CAR_WIDTH) {
        redCarX += 5;
    }

    if (isBlueCarMovingLeft && blueCarX > 0) {
        blueCarX -= 5;
    } else if (isBlueCarMovingRight && blueCarX < TRACK_WIDTH - CAR_WIDTH) {
        blueCarX += 5;
    }
}

private void checkFinishLine() {
    if (redCarX >= FINISH_LINE && blueCarX >= FINISH_LINE) {
        isGameRunning = false;
        isGameFinished = true;
        if (redCarX >= blueCarX) {
            winnerMessage = "Red car wins!";
        } else {
            winnerMessage = "Blue car wins!";
        }
    }
}

public void paint(Graphics g) {
    g.setColor(Color.gray);
    g.fillRect(0, 0, WINDOW_WIDTH, WINDOW_HEIGHT);
    g.setColor(Color.black);
    g.fillRect((WINDOW_WIDTH - TRACK_WIDTH) / 2, (WINDOW_HEIGHT -
TRACK_HEIGHT) / 2, TRACK_WIDTH, TRACK_HEIGHT);
    g.setColor(Color.red);
    g.fillRect(redCarX, redCarY, CAR_WIDTH, CAR_HEIGHT);
    g.setColor(Color.blue);
    g.fillRect(blueCarX, blueCarY, CAR_WIDTH, CAR_HEIGHT);

    if (isGameFinished) {
        g.setColor(Color.white);
        g.setFont(new Font("Arial", Font.BOLD, 36));
    }
}

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        int messageWidth = g.getFontMetrics().stringWidth(winnerMessage);
        g.drawString(winnerMessage, (WINDOW_WIDTH - messageWidth) / 2,
WINDOW_HEIGHT / 2);

        // Celebration animation
        for (int i = 0; i < 10; i++) {
            g.setColor(new Color((int)(Math.random() * 256),
(int)(Math.random() * 256), (int)(Math.random() * 256)));
            int x = (int)(Math.random() * WINDOW_WIDTH);
            int y = (int)(Math.random() * WINDOW_HEIGHT);
            int size = (int)(Math.random() * 50) + 10;
            g.fillOval(x, y, size, size);
        }
    }

    public void keyPressed(KeyEvent e) {
        int keyCode = e.getKeyCode();
        if (keyCode == KeyEvent.VK_LEFT) {
            isRedCarMovingLeft = true;
        } else if (keyCode == KeyEvent.VK_RIGHT) {
            isRedCarMovingRight = true;
        } else if (keyCode == KeyEvent.VK_A) {
            isBlueCarMovingLeft = true;
        } else if (keyCode == KeyEvent.VK_D) {
            isBlueCarMovingRight = true;
        }
    }

    public void keyReleased(KeyEvent e) {
        int keyCode = e.getKeyCode();
        if (keyCode == KeyEvent.VK_LEFT) {
            isRedCarMovingLeft = false;
        } else if (keyCode == KeyEvent.VK_RIGHT) {
            isRedCarMovingRight = false;
        } else if (keyCode == KeyEvent.VK_A) {
            isBlueCarMovingLeft = false;
        } else if (keyCode == KeyEvent.VK_D) {
            isBlueCarMovingRight = false;
        }
    }

    public void keyTyped(KeyEvent e) {
        // Do nothing
    }

```

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}
```

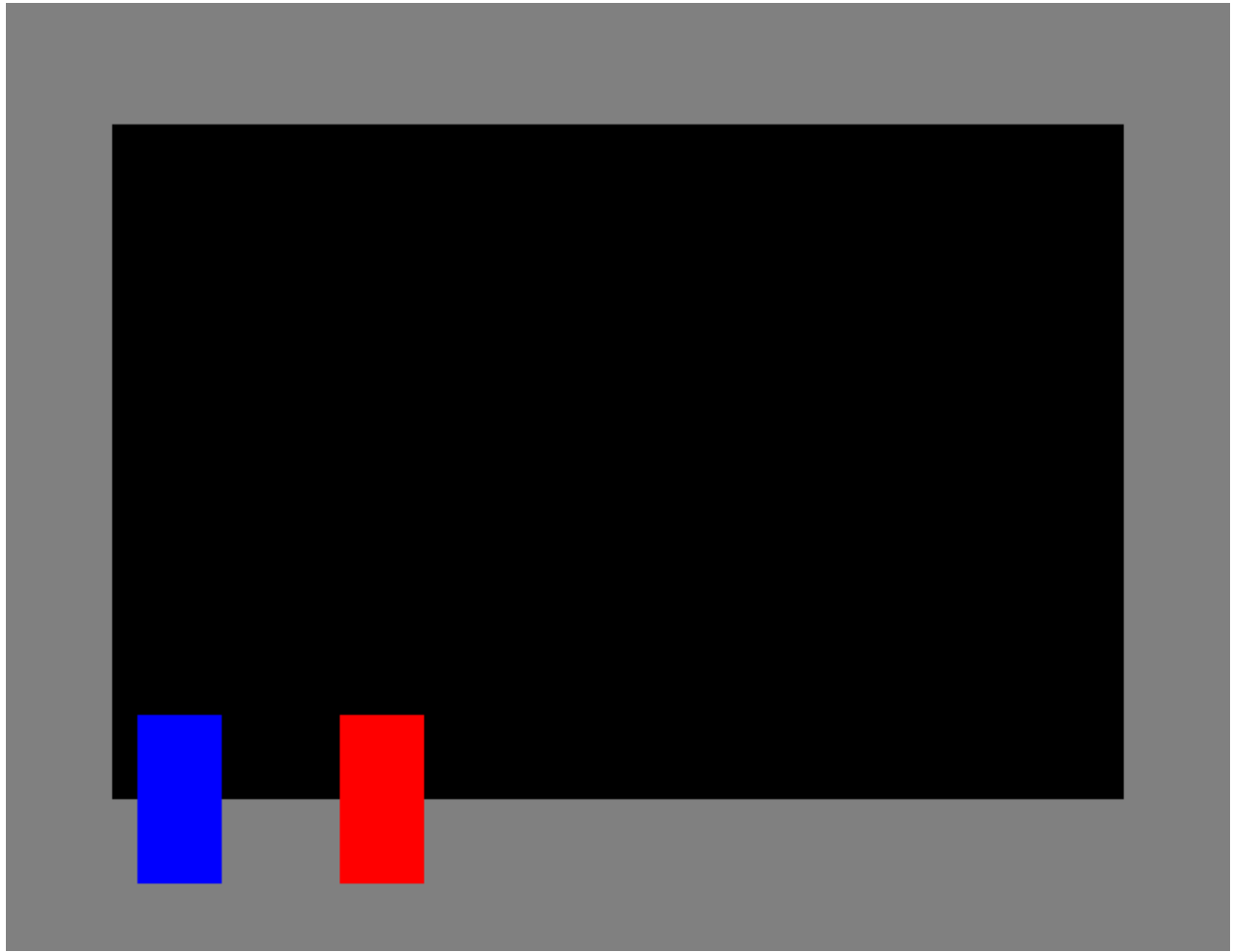
HTML CODE:

```
<!DOCTYPE html>
<html>
<head>
  <title>Car Racing Game</title>
</head>
<body>
  <h1>Car Racing Game</h1>
  <p>Use the left and right arrow keys to control the red car, and the A and D
keys to control the blue car.</p>
  <p>The first car to reach the finish line wins!</p>
  <hr>
  <applet code="CarRacingGame.class" width="800" height="600">
    Your browser does not support Java Applets.
  </applet>

  <script>
    // Add any additional JavaScript code if needed
  </script>
</body>
</html>
```

OUTPUT;

BLUE CAR IS WINNER.



RED CAR IS WINNER.

